

Polydopamine functionalised ceria-zirconia nanoparticles embedded water-borne epoxy nanocomposite for anti-biofouling coatings

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ABSTRACT

Biofouling, the accumulation of microorganisms, plants, algae on wet surfaces is one of the major issues adversely affecting the overall hydrodynamic performance of the marine vessels. Ceria (CeO_2) nanoparticles (NPs) are effectively used as anti-biofouling agent to prevent the deterioration of steel structures, due to their excellent redox capacity. Various approaches are being investigated to enhance the antifouling activity of ceria NPs. Here, we report the development of novel polydopamine (PDA) functionalised ceria-zirconia nanoparticles filled water-borne epoxy nanocomposite coating to prevent the microbial-induced corrosion of mild steel. Ceria NPs were functionalised with PDA to enhance the dispersibility and improve their ability to resist biofouling in water-borne epoxy resin coatings against microbial species. As the anti-biofouling activity of ceria depends on their oxygen storage capacity, zirconium was incorporated to create a defective crystal structure with more active oxygen storage and release sites. Ceria and ceria-zirconia NPs were synthesised by precipitation method and functionalised with PDA. The functionalisation of ceria-zirconia (CZ) NPs was confirmed by Fourier Transform Infrared (FTIR) spectroscopy, X-ray Diffraction (XRD), Confocal Raman spectroscopy and X-ray Photon spectroscopy (XPS). Thermogravimetric analysis (TGA) shows that incorporation of zirconium increases the amount of oxygen vacancies in the crystal lattice thereby enhancing ceria's redox potential. Anti-biofouling and anti-biocorrosion properties of the coatings were explored against different microbial strains. The antibacterial tests show that colony-forming units (CFU) of PDA functionalized ceria-zirconia epoxy (EPCZ) nanocomposites were suppressed by 93 % and 87 % in the case of *Staphylococcus aureus* (*S. aureus*) and *Pseudomonas aeruginosa* (*P. aeruginosa*) bacteria, respectively. Biofilm inhibition studies show that EPCZ nanocomposite coatings have higher biofilm inhibition efficiency against gram-negative bacteria (96.37 %) than gram-positive bacteria (62.67 %) due to the differences in their morphology. The practical applicability of the nanocomposite coatings was studied using cultured seawater consortia, and the results reveal that PCZ nanoparticles provide superior biofilm inhibition. Electrochemical impedance spectroscopy (EIS) reveals that the EPCZ coatings exhibit significant enhancement in charge transfer resistance and coating resistance. The synergistic effect of PDA functionalisation and zirconium incorporation in ceria leads to the exceptional biocorrosion resistance in corrosive bacterial environments.

1. Introduction

Biofouling refers to the undesirable colonisation and growth of microbes, invertebrates, and microalgae on artificial surfaces, resulting in their bio-deterioration. The microorganisms adhere to the surfaces and build a biofilm layer, allowing the growth of other aquatic species, like diatoms and algae, which pose a significant threat to materials in marine environments [1,2]. In the maritime environment, this chain of events considerably impacts various surfaces, including pipelines, water intake

systems, probes and sensors, ship hulls, building materials, and filters. In heat exchangers, biofouling reduces the heat transfer, in addition to the corrosion of the metal structures [3]. Marine biofouling-induced frictional drag tends to reduce ship speed by 10 %; as a result, a severely fouled ship will burn up to 40 % more fuel, resulting in an economic burden. Thus, employing anti-biofouling coatings can significantly lower operating costs, particularly for naval vessels, resulting in cost savings [4,5]. Various methods to control surface fouling include using biocides, toxic coatings, etc. Antifouling paints based on organic tin

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derivatives, mainly tributyl tin, were widely used in the 1970s; however, due to their toxicity towards marine organisms, these materials were banned [4]. Coating systems made from alkyd resins, epoxy resins, and other materials are widely used to prevent the corrosion and fouling of the surfaces [6,7]. Due to the low emissions of volatile organic compounds and high adhesive nature, water-borne epoxy coatings are environmentally safe and have exceptional barriers, hence having corrosion protection properties [3]. However, epoxy coatings are susceptible to long-term corrosion due to the transport of oxygen and aggressive species from the electrolyte to the metallic substrate through the pores in the coating layers [8,9]. One potential approach for overcoming this limitation is the incorporation of nanomaterials in the coatings composed of water-borne epoxy polymers [10–12]. The antibacterial properties of different metals and metal oxide nanoparticles with enhanced stability, such as Silver, Copper, Zinc oxide (ZnO), Magnesium oxide (MgO), Titanium oxide (TiO₂), etc. are extensively studied. Despite their outstanding antibacterial properties, many nanoparticles, such as silver, are harmful to marine life, even at low concentrations [13].

Recent studies reported the anti-biofouling activity of the rare earth metal CeO₂ nanoparticles. Unlike the requirement of photoactivation by UV light or visible light for the functioning of TiO₂ and ZnO NPs, CeO₂ NPs exhibit antibacterial activity without any external activation [14]. The exceptional antibacterial activity of CeO₂ NPs is attributed to its ability to produce and migrate oxygen vacancies, facilitated by an autoregenerative cycle that allows for reversible transition between the two valence states (Ce⁴⁺ ↔ Ce³⁺) [14,15]. This process leads to an oxy-reduction reaction that generates reactive oxygen species (ROS), including hydroxyl radicals (OH•) and singlet oxygen (O₂•). ROS causes the lipids and proteins in the plasma membrane of bacterial cells to undergo oxidative stress, increasing membrane permeability and cell mortality [15,16]. CeO₂ NPs have a high oxygen storage capacity (OSC) because of unpaired oxygen vacancies, a critical attribute for numerous applications such as corrosion protection, oxygen storage, UV absorbers, catalysis, and so on [17,18]. Another mechanism responsible for cell death is electrostatic interaction between positively charged CeO₂ nanoparticles and negatively charged bacteria. The nanoparticles are adsorbed onto the bacterial membrane, binds with mesosomes to disrupt cellular respiration, nutrition transport, DNA replication, and finally induce cell death while increasing the membrane permeability [14,19,20]. CeO₂ alone does not demonstrate significant antifouling activity in practical applications due to its low solubility and dispersibility in the coating matrix. This can be overcome by the surface modification of these materials with polymers such as polydopamine, which improves their stability and interaction with the coating matrix. The catechol OH groups in polydopamine exhibit a high affinity for metal coordination, which improves adhesion between the substrate and the coating [21–23]. Extensive works have been reported on the use of nanomaterials modified with PDA to improve their dispersibility [24,25].

One of the methods to enhance the OSC of CeO₂ is doping. Previous studies investigate that introducing zirconium as a dopant significantly improves the OSC and thermal stability of CeO₂. The transition of Ce⁴⁺ → Ce³⁺ results in an undesirable lattice strain in undoped ceria due to the size difference between the ions. When zirconium is added to the crystal structure of ceria, the Zr⁴⁺ cations, which are of smaller size, get accommodated into the defect sites of the crystal and reduce the lattice strain, thereby enhancing the thermal stability [26,27]. In addition, incorporating Zr into cubic CeO₂ results in a defective crystal structure with more active oxygen storage and release sites (e.g., weakly bound oxygen, delocalised oxygen vacancies and even oxygen interstitials) [27,28]. Joseph et al. reported that zirconium-doped ceria nanoparticles in water-borne epoxy polymer coatings serve as an anticorrosion pigment on mild steel [17]. It is reported that incorporation of Zr in CeO₂ greatly enhances oxygen storing and releasing capacities, improving the corrosion resistance of water-borne epoxy coatings.

Numerous studies have reported the enhanced thermal stability, corrosion resistance behaviour and OSC of CZ NPs [29–31]. The antibacterial efficiency of ceria-zirconia nanoparticles has also been evaluated [32,33]. However, the anti-biofouling capabilities of these nanoparticles are yet to be explored in detail, which is the novelty of the present work.

Due to the potential synergistic effect that Zr and Ce species may have on bacterial inhibition, it is prudent to conduct more research on this composite coating for the development of a facile and eco-friendly coating to suppress the biofouling on the surfaces, submerged in seawater. The present work mainly focuses on developing a novel EPCZ polymer nanocomposite film and investigating its anti-biofouling and biocorrosion properties on a mild steel substrate. This work also optimises the chemical composition of Ce_xZr_{1-x}O₂ with varying Zr⁴⁺ molar ratios and ceria/dopamine hydrochloride ratios required to improve the formation of active sites for anti-biofouling action. To investigate the ability of water-borne epoxy nanocomposite coatings to resist biofouling, experiments were conducted using gram +ve and gram -ve bacteria. The bacterial strains were grown in a nutrient broth, and the coatings were immersed in the culture medium to study their antibiofilm properties. Additionally, bioelectrochemical analyses revealed the anti-biocorrosion properties of various epoxy coatings. The physicochemical mechanisms underlying the improved antifouling potential are also discussed.

2. Experimental section

2.1. Materials

Reagents cerium (III) nitrate hexahydrate (Ce(NO₃)₃·6H₂O), zirconyl chloride octahydrate (ZrOCl₂·8H₂O), and ammonium hydroxide (NH₄OH, 25 % NH₃) was procured from Merck and dopamine hydrochloride was procured from Sigma Aldrich. Formaldehyde (37–41 %) and osmium tetroxide (4 % aqueous) were procured from Thermo Fisher Scientific India. Bacterial strains *P. aeruginosa* (MTCC 4240), *Escherichia coli* (*E. coli*) (MTCC 443) and *S. aureus* (MTCC 96) were supplied by the National Centre for Cell Science, Pune, India. Bacterial culture medium, Nutrient broth (NB), and phosphate buffer solution with a pH of 7.4 were purchased from Himedia, India. All culture media underwent sterilisation at a temperature of 121 °C, under a pressure of 15 psi, for 20 min before use. All commercial reagents utilised in this study were of analytical grade.

In the present work, the substrate material used was mild steel (0.025 wt% C, 0.194 wt% Mn, 0.013 wt% Si, 0.005 wt% S, 0.008 wt% P, 0.056 wt% Al, and 99.639 wt% Fe). For anti-biofouling experiments, the dimensions of the mild steel used were 1 cm × 1 cm, and for bio-corrosion investigations, the dimension was 2 cm × 7.5 cm.

2.2. Synthesis of ceria nanoparticles and ceria-zirconia nanoparticles

A straightforward precipitation approach was adopted to synthesise ceria nanoparticles. The synthesis process involved using Ce(NO₃)₃·6H₂O as the raw material and NH₄OH as the precipitating agent. Then, 3 M NH₄OH solution was mixed with precursor dropwise with continuous stirring until a yellow precipitate was formed. After mixing, the solution was centrifuged at 4000 rpm for 10 min. Multiple washes with deionised water followed this process to eliminate organic contaminants. The remaining residue was subjected to drying in a laboratory oven set at 60 °C and left overnight, resulting in the formation of cerium hydroxide (Ce(OH)₃) powder. Afterwards, the powder was calcined at 500 °C for 3 h to yield ceria nanoparticles.

Co-precipitation was used to synthesise CZ NPs. The different ratios of ceria to zirconia, namely Ce_{0.8}Zr_{0.2}O₂, Ce_{0.6}Zr_{0.4}O₂ and Ce_{0.5}Zr_{0.5}O₂, were obtained by varying the concentrations of Ce(NO₃)₃·6H₂O and ZrOCl₂·8H₂O as precursor material. Aqueous ammonia was the reducing agent in this process [34]. The aqueous solutions of Ce(NO₃)₃·6H₂O and ZrOCl₂·8H₂O (as mentioned in Table 1) were agitated for 15 min at

Table 1

Sample code and respective components distribution in nanoparticles ("E" will be added as a prefix for epoxy coatings with the nanoparticles).

Sample code	wt% of ceria	wt% of zirconium	wt% of PDA
C	~100	0	0
CZ2	~80	~20	0
CZ4	~60	~40	0
CZ5	~50	~50	0
P	0	0	100
PC4	~40	0	~60
PC5	~50	0	~50
PC8	~80	0	~20
PCZ	~48	~32	~20

ambient temperature. The solution was gradually diluted by adding 3 M aqueous ammonia until the pH is more than 10, while stirring the solution vigorously. The yellow precipitate obtained was centrifuged at 4000 rpm for 10 mins, dried and calcined at 500 °C for 3 h to get CZ NPs.

2.3. Functionalisation of nanoparticles with polydopamine

A chemical oxidative polymerisation reaction was used to functionalise CeO₂ and CZ NPs with PDA [35]. First, the CZ NPs (40, 50, and 80 mg) were dispersed in 50 mM Tris-HCL solution with a pH of 8.5 and sonicated for 30 min. Subsequently, dopamine hydrochloride (60 mg, 50 mg, and 20 mg) was added to the mixture and agitated for 24 h. The resulting solution was collected and centrifuged at 4000 rpm for 10 min. The sediment obtained was washed repeatedly with water and ethanol solution. Finally, the obtained PDA-functionalized ceria NPs were dried at 80 °C for 24 h. Pristine PDA and PCZ NPs were also synthesised using a similar procedure. Fig. 1 illustrates a diagrammatic representation of the synthesis procedure.

2.4. Preparation of nanocomposite coatings

Nanocomposites (C, CZ4, PC8 and PCZ) were prepared by dispersing synthesised nanoparticles in epoxy resin (E). The details of the samples prepared for the experiments are shown in Table 1. Initially, 0.5 wt% of nanoparticles was dispersed in deionised water and subjected to sonication for 10 min. Water-borne epoxy resin was added to the solutions and sonicated for 30 min to obtain homogeneous dispersions of C, CZ4, PC8 and PCZ nanoparticles within the epoxy matrix. The mild steel substrate (2 cm × 2 cm) was initially polished using silicon carbide papers of different mesh sizes, ranging from 300 to 1500 mesh, followed by ultrasonic washing with acetone and drying. The nanocomposites in pure epoxy were coated onto the mild steel substrate using a spin coater (Holmarc, Model: HO-TS-O5S) with a rotational speed of 200 rpm for 30 s. Spin-coated samples were cured at 90 °C for 2 h, followed by 110 °C for 2 h, and finally at 150 °C for 2 h. The thickness of the cured films was measured using a Surfex® pro gauge (PHYNIX Sensortechnik GmbH, Germany). A cross-hatch adhesion tester (Elcometer 107) was used to test the different coatings-pull-off adhesion under the ASTM D 3359-B protocol. A 2D surface profilometer (Mitutoyo SJ-410) was used to measure the surface roughness of the nanocomposite coatings.

2.5. Characterization of nanoparticles and nanocomposite coatings

The morphology and crystallinity of CeO₂ and CZ NPs were examined using a high-resolution transmission electron microscope (HRTEM) equipped with selected-area electron diffraction (SAED). Specifically, the HRTEM instrument used in this study was the Tecnai F20 model, operating at an accelerating voltage of 200 kV. HRTEM allows for the visualisation of the nanoscale morphology and the examination of crystal structures at high resolution. SAED provides valuable

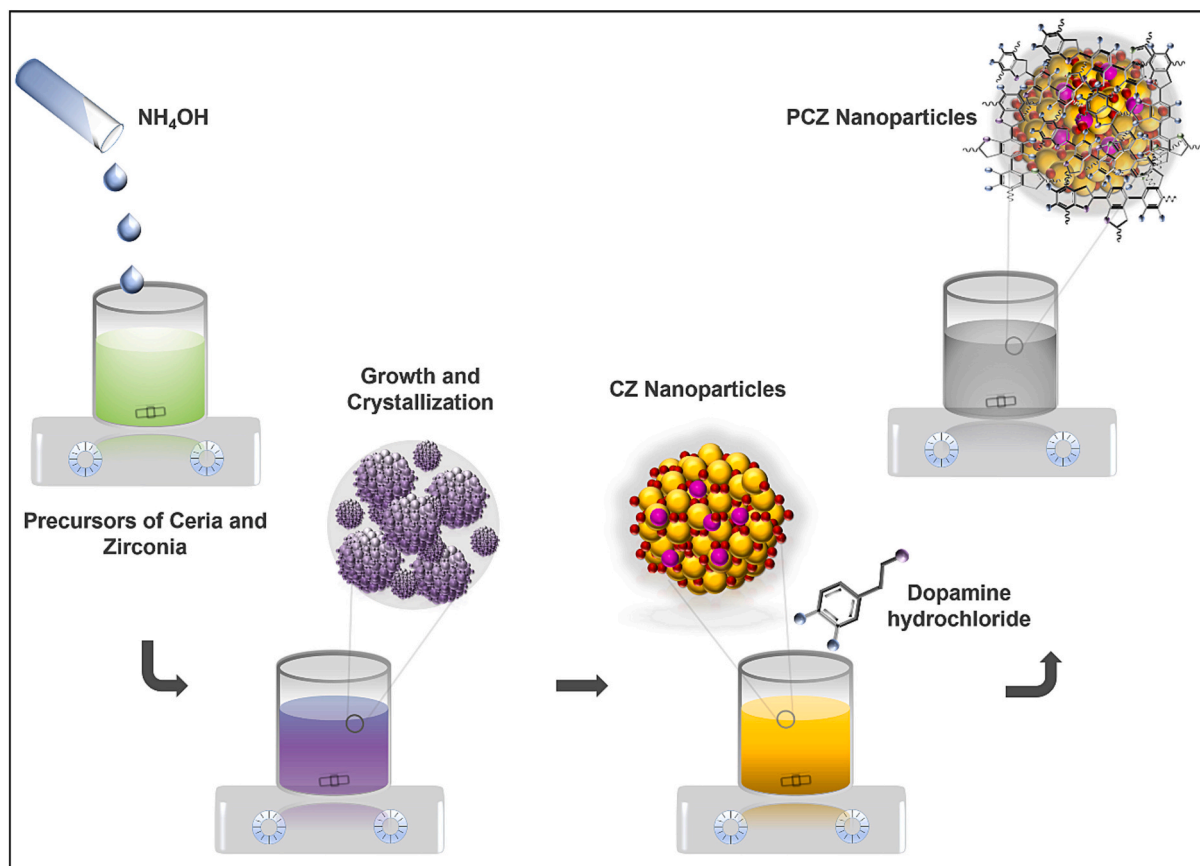


Fig. 1. Synthesis route for PCZ nanoparticles - schematic.

information about the crystallographic orientations and phases present in the nanoparticles. X-ray diffraction (XRD) and Raman spectroscopy were performed to analyse the oxygen vacancy concentration and the lattice strain in CZ nanoparticles (CZ2, CZ4, and CZ5). The XRD studies used a Malvern PANalytical X'Pert3 powder diffractometer equipped with Cu-K α radiation ($\lambda = 1.540 \text{ \AA}$). Raman spectroscopy measurements were conducted using a confocal Raman microscope (Labram HR Evolution, Horiba, France) in the spectral range of 200–1000 cm^{-1} with a 532 nm laser as the excitation source.

The functionalisation of nanoparticles was studied at ambient temperature (25 °C) by ATR method using FTIR spectrometer (PerkinElmer (USA)). In every instance, 32 scans having a resolution of 4 cm^{-1} were employed in a spectral range of 400–4000 cm^{-1} . The FTIR spectra of P, PC8, and PCZ nanoparticles were individually recorded to confirm the functionalisation of PDA on ceria nanoparticles. To ascertain the surface elemental composition and valency of PCZ nanoparticles, X-ray photoelectron spectroscopy (XPS) analysis (Omicron Nanotechnology) was performed using a Mg K α X-ray source with 1253.6 eV of excitation energy.

Thermogravimetric analysis (Waters TA instruments, TGA Q50) was used to study the OSC of C and CZ4 nanoparticles [36]. This method involves multiple cycles of heating and cooling samples at a ramp rate of 10 °C/min. In the first cycle, the sample gets heated to 750 °C in a nitrogen environment and then cooled to 150 °C in an oxygen environment. In the second cycle, the specimen was again heated to 750 °C in atmosphere containing nitrogen, and then cooled to ambient temperature in an atmosphere containing oxygen. The mass change during 2nd heating is accounted for in the OSC calculation. Nanocomposite coatings were subjected to Field Emission Scanning Electron Microscopy (FESEM) and Energy dispersive X-ray spectroscopy (EDAX) to study their surface morphology and elemental composition (ZEISS Gemini SEM 300).

2.6. Antibacterial activity of nanoparticles

The antimicrobial efficiency of synthesised nanoparticles and nanocomposites was evaluated against three microbial strains: *S. aureus* (gram +ve), *P. aeruginosa*, and *E. coli* (gram –ve) by plate count method [37]. 20 μL of bacterial cultures (with absorbance adjusted to 0.1) were added to each nanoparticle (C, PC4, PC5, PC8, CZ2, CZ4, CZ5 and PCZ) and the nanocomposites (E, EC, EPC8, ECZ4, and EPCZ) with a concentration of 10 mg/mL and were incubated at 37 °C for 24 h. Following the specified incubation period, 100 μL of each composite containing microorganisms was serially diluted using sterile deionised water. 200 μL of the diluted microbial suspension was spread evenly onto nutrient agar plates. These plates were then incubated at 37 °C for 24 h. After incubation, the number of CFUs on each plate was recorded. This count of CFUs allowed for determining the bacterial growth rate in each sample, providing insights into the antimicrobial effectiveness of the tested nanoparticles and nanocomposites.

2.7. Biofilm analysis by crystal violet staining method

The quantification of biofilm formation by microbial strain was carried out using crystal violet staining method, for a concentration of 1 mg/mL [33,38,39]. In brief, an overnight-grown bacterial cell culture was treated with varying concentrations of C, PC8, CZ4, and PCZ (100 to 400 $\mu\text{g/mL}$). This mixture was then transferred to a 96-well microtiter plate containing NB and incubated for 72 h at 37 °C without shaking. After removing the suspended bacteria, the surface-attached biofilm cells underwent a series of washing steps with sterile distilled water. Following washing, adherent cells remaining on the 96-well microtiter plate were stained with crystal violet and left to incubate at room temperature for 20 min to allow for proper staining. The stained cells were then washed thrice with water to eliminate excess dye. To quantify the biofilm, the stained cells underwent dissolution in 95 % ethanol. The

resultant solution's optical density (OD) was evaluated at a wavelength of 600 nm using a microplate reader. Each experiment trial was conducted thrice to ensure reliable results and statistical significance. The efficiency of nanomaterials to inhibit biofilm formation can be quantitatively calculated using the following equation [40];

$$\% \text{biofilm inhibition} = \frac{(\text{Positive Control OD}_{600 \text{ nm}} - \text{Sample OD}_{600 \text{ nm}})}{\text{Positive Control OD}_{600 \text{ nm}}} \times 100 \quad (1)$$

2.8. Anti-biofouling properties of nanocomposite coatings

Mild steel substrates spin-coated with E, EC, EPC8, ECZ4, and EPCZ, and an uncoated substrate (control) were immersed in bacterial culture. After five days of exposure, the samples were treated, and bacterial adhesion profiles were analysed using FESEM. The microscopic images would provide a good understanding of the anti-biofouling capability of each sample. To assess the practical applicability of the developed coatings, seawater was collected from Calicut Harbour in Kozhikode, Kerala, India, and used as the inoculum for the bacteria culture preparation. Control, E, and EPCZ coatings were submerged in a culture medium inoculated with seawater to comprehend the anti-biofouling activity of the coatings. The immersion period lasted 15 days, during which the coatings interacted with the bacterial population in the seawater culture. The coatings were then treated after incubation, and the extent of bacterial adherence and biofilm formation rate on different coating formulations were analysed using SEM images.

2.9. Anti-biocorrosion activity of nanocomposite coatings

The anti-biocorrosive activity of the nanocomposite coatings was studied using EIS. The bioelectrochemical measurements were conducted using an electrochemical workstation, specifically the Metrohm Autolab PGSTAT 128N instrument. For the experiments, 50 mL of NB containing bacterial strains was prepared and incubated for 24 h at 37 °C. A three-electrode system was utilised, with the nanocomposite-coated sample of 1 cm^2 exposed surface area, as the working electrode. Graphite and Ag/AgCl electrodes were employed as the auxiliary and reference electrodes, respectively. EIS investigations were done at the open circuit potential (OCP) and a sinusoidal signal with an amplitude of 10 mV was applied over frequency range 100 kHz to 10 mHz. Multiple measurements were taken at different time intervals, specifically at 12, 48 and 84 h of exposure. EIS data was fitted using an equivalent circuit, and the EC lab software (EC-Lab V10.40) was utilised to obtain the fitting parameters.

3. Results and discussions

3.1. Characterization of nanoparticles and nanocomposite coatings

HR-TEM was used to examine the morphology of CZ4 and PCZ nanoparticles, as shown in Fig. 2. Fig. 2(a) clearly illustrates that CZ4 nanoparticles are spherical. From the high-resolution image of CZ4 nanoparticles, as shown in Fig. 2(b), the distance between the adjacent lattice planes is 0.204 nm, which can be assigned to the (220) crystallographic planes of CZ4 (ICDD 38-1439) crystals. The plane was assigned based on similarities in d-spacing between XRD and SAED. The crystalline nature of CZ4 nanoparticles was proven by the presence of diffraction rings in the SAED pattern, as given in Fig. 2(c). The SAED pattern exhibited seven continuous rings corresponding to the (111), (200), (220), (311), (400), (331), and (420) planes, indicating a cubic crystal structure. Fig. 2(d) shows the TEM image of PCZ nanoparticles, which confirms their spherical morphology and demonstrates a reduction in agglomeration compared to CZ nanoparticles. The SAED pattern displayed in Fig. 2(e) confirmed that the particle is polycrystalline, as evidenced by diffraction rings. The size distribution of PCZ NPs is

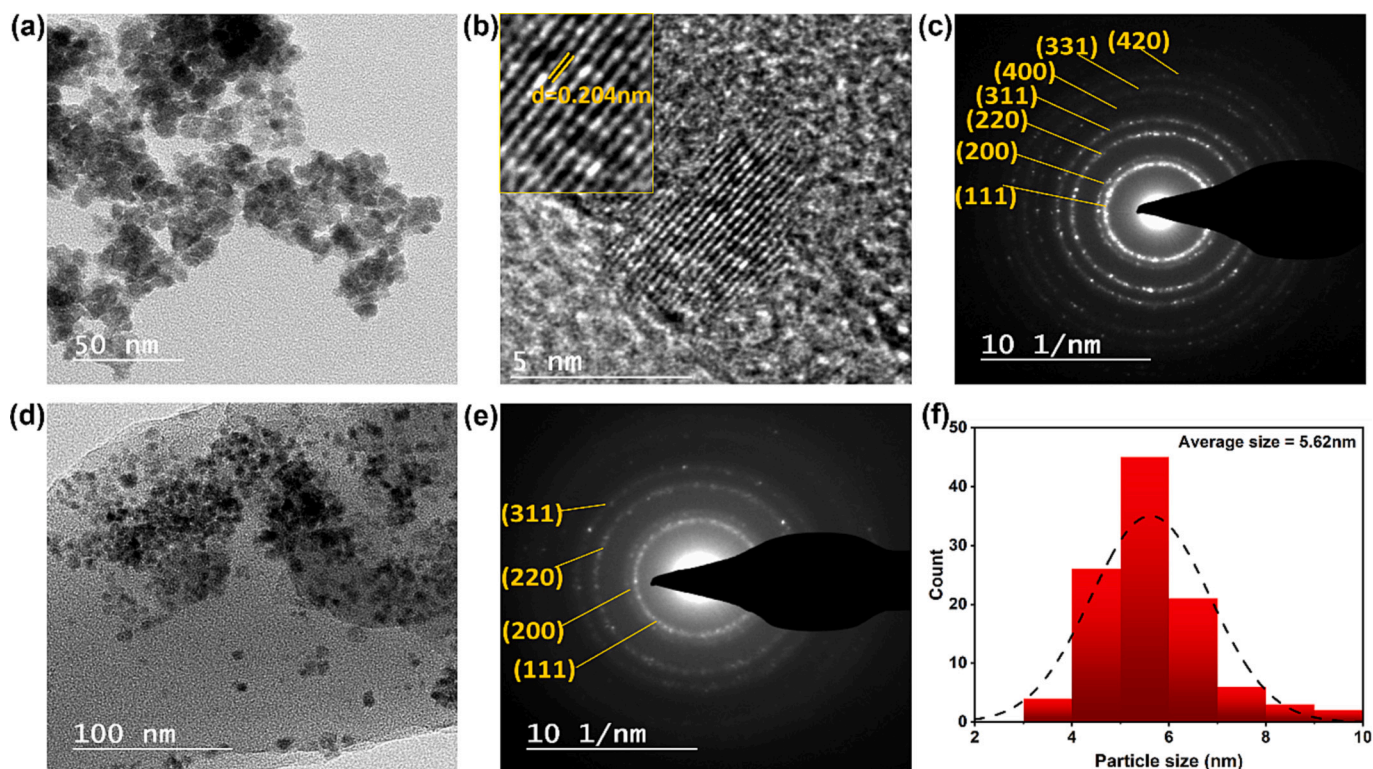


Fig. 2. (a) TEM image of CZ4 nanoparticles, (b) HR-TEM image of CZ4 nanoparticles and the inset shows interplanar spacing ($d = 0.204$ nm), (c) SAED pattern of CZ4 nanoparticles. (d) TEM image of PCZ nanoparticles. (e) SAED pattern of PCZ nanoparticles. (f) Particle size distribution of PCZ nanoparticles (obtained from TEM).

depicted in Fig. 2(f), revealing an average size of the particle approximately 5.62 nm.

The XRD spectra of pure ceria and CZ NPs are depicted in Fig. 3(a). The obtained peaks at specific 2θ values are as follows: 28.51° (111), 33.04° (200), 47.45° (220), 56.34° (311), 69.42° (400), and 79.01° (420). The aforementioned peaks represent the cubic phase structure of ceria, which matches the standard JCPDS data card (PDF-ICDD 43-1002) [41]. The XRD pattern of CZ NPs indicates that the 2θ values of the peaks (111), (200), (220), (311), (400), and (420) were broadened and slightly shifted to the higher diffraction angles, indicating Zr addition in CeO_2 . Notably, no additional peaks corresponding to $\text{ZrO}_2/\text{Zr}(\text{OH})_2$ phases were observed, indicating their absence in the nanoparticles. The smaller ionic radius of Zr^{4+} (0.84 Å) compared to Ce^{4+} (0.97 Å) leads to the integration of zirconium ions into the crystal unit of ceria, reducing cell parameters and lattice shrinkage. CZ-NPs exhibit a phase transition from cubic (CZ2 and CZ4) to tetragonal (CZ5) crystal structure. All reflections become less intense and broader as the zirconium content of the nanoparticle increases, implying that the inclusion of Zr into the ceria lattice causes a reduction in crystallite size and an increase in lattice strain. Table 2 shows the nanoparticles' average crystallite sizes, deduced using the Debye-Scherrer equation based on the corresponding diffraction peaks. As the concentration of Zr increases, the crystallite size decreases, with CZ5 nanoparticles exhibiting the smallest crystallite size. Therefore, it is evident that incorporation of Zr into the ceria lattice inhibits crystalline growth. Furthermore, adding Zr to CeO_2 leads to the contraction of the CeO_2 lattice, as evident from the lattice parameter values presented in Table 2. The lattice parameter decreases from 5.4136 Å in pure ceria to 5.334 Å in the CZ nanoparticles. An ionic size mismatch between Zr and Ce could cause lattice contraction. The difference in the ionic radii generates local disruptions in the lattice structure, leading to strain that impedes the nucleation rate. Consequently, the particle size is reduced due to this strain-induced effect [17].

Williamson-Hall (W—H) analysis examines the broadening peak width as a function of 2θ obtained from XRD data to determine the relationship between crystallite size (D) and lattice strain (ϵ) [42]. The following is the W—H equation:

$$\beta \cos \theta = \frac{K\lambda}{D_v} + 4\epsilon \sin \theta \quad (2)$$

where ϵ is the lattice strain, D_v is the average crystallite size, λ is the incident X-ray wavelength, β is the full width at half maximum (FWHM) of the diffraction peaks, θ is the diffraction angle, and K is the shape factor (~ 0.89). To determine the strain and crystallite size, $4\epsilon \sin \theta$ can be plotted along the X-axis and $\beta_{hkl} \cos \theta$ along the Y-axis, as shown in Fig. S1. The slope of the linear fit represents the effective strain, while the intercept represents the effective crystallite size. The measured crystallite size and lattice strain obtained from this method are shown in Table 2. Comparing the average crystallite sizes derived using Scherrer and W—H methods, the values are similar, ranging from 2.83 to 6.42 nm, suggesting a close agreement between the two approaches in determining the average crystallite size. Analysis of lattice strain demonstrates that an increase in zirconium concentration leads to higher lattice strain. This increment in lattice strain can be attributed to two factors: a decrease in particle size and the formation of oxygen vacancies. The addition of zirconium to the ceria lattice results in a reduction in particle size. As a consequence, the local symmetry of the ceria lattice is distorted due to the size difference between Ce^{4+} and Ce^{3+} ions. This distortion induces strain within the lattice structure [43].

The oxygen vacancy concentration in ceria and CZ NPs was determined using the lattice parameters determined by XRD studies. Each oxygen atom expelled from the lattice leaves a vacancy $\text{V}_\text{O}^{\bullet\bullet}$, forming two Ce^{3+} cations by giving electrons to two Ce^{4+} cations. Each sample's total oxygen vacancy concentration is calculated using Zhou and Huebner's equation [44]

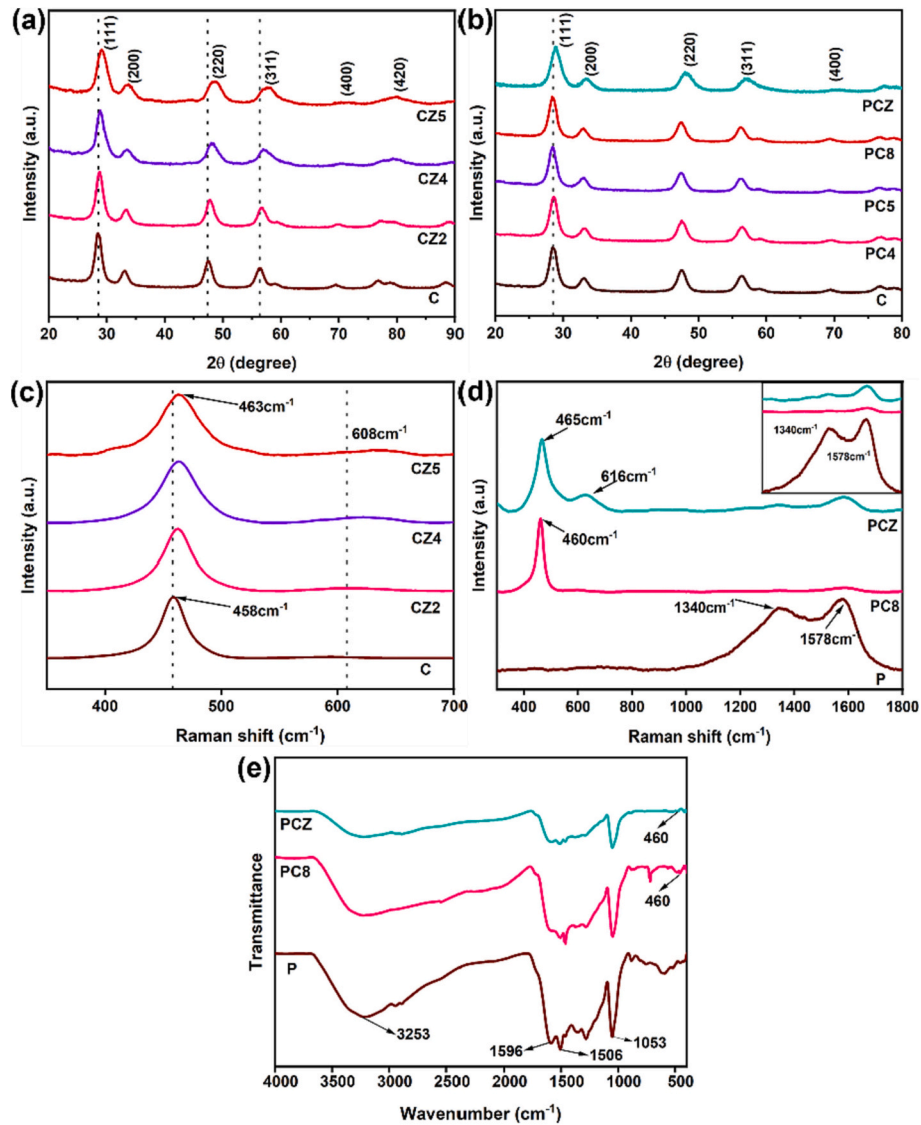


Fig. 3. (a) XRD patterns of pure ceria and CZ nanoparticles. (b) XRD pattern of ceria and different concentrations of PDA functionalized ceria nanoparticles and PCZ nanoparticles. Raman spectra of (a) ceria and CZ NPs. (b) PDA functionalized nanoparticles. FTIR spectra of P, PC8 and PCZ.

Table 2

Deduced parameters from the XRD Data.

Sample	Structure	Crystallite size (nm)		Lattice parameters			Lattice strain	Oxygen vacancy concentration (cm ⁻³)
		Scherrer method	W-H method	a (Å)	b (Å)	c (Å)		
C	Cubic	6.42	3.84	5.413	5.413	5.413	-0.0068	1.023 × 10 ¹⁸
CZ2	Cubic	6.35	3.27	5.373	5.373	5.373	-0.0091	9.447 × 10 ¹⁸
CZ4	Cubic	5.30	2.99	5.343	5.343	5.343	-0.0052	1.4 × 10 ¹⁹
CZ5	Tetragonal	5.28	2.83	3.744	3.744	5.334	-0.0053	

$$\frac{\sqrt{3}}{4}(a' - a_0) = c \left[r_{\text{Ce}^{3+}} - r_{\text{Ce}^{4+}} + \frac{1}{4}(r_{\text{V}_\bullet} - r_{\text{O}^{2-}}) \right] \quad (3)$$

where $r_{\text{Ce}^{3+}}$ is the radius of Ce^{3+} (0.1283 nm), $r_{\text{Ce}^{4+}}$ is the radius of Ce^{4+} (0.1098 nm), $r_{\text{V}_\bullet}$ is the radius of oxygen vacancy (0.138 nm), $r_{\text{O}^{2-}}$ is the radius of O^{2-} (0.124 nm), a' and a_0 is the lattice parameter of the synthesised ceria and CZ NPs respectively. The parameter c in the above equation provided is equal to the $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio and is related to vacancy concentration through the equation [45],

$$V_\bullet = \text{O}^{2-} \times \frac{c}{4} \quad (4)$$

Considering that both Ce^{3+} and Ce^{4+} have the same coordination number of eight O^{2-} ions, we can write $\text{V}_\bullet = 2c$ (assuming $\text{O}^{2-} = 8$). By utilising the above equations and parameters, the calculated values of $\text{V}_\bullet$ (oxygen vacancy concentration) for each sample were recorded in Table 2. An increase in oxygen vacancy concentration accompanies the increment in zirconium concentration. The highest oxygen vacancy concentration was observed for CZ4, with a value of $1.4 \times 10^{19} \text{ cm}^{-3}$. Zr incorporation causes lattice deformation, which increases oxygen defect density.

Fig. 3(b) depicts the XRD spectra of ceria and different concentrations of PCZ NPs. The diffraction peaks observed at 28.49° , 33.1° ,

47.47°, 56.28° and 69.36° correspond to the (111), (200), (220), (311), and (400) lattice planes of ceria, respectively. These peaks indicate cubic fluorite structure in both ceria and PCZ NPs. The XRD analysis confirmed the functionalisation of PDA. However, from the XRD spectra, it was found that the characteristic peaks of PCZ NPs follow a similar pattern to that of ceria without any noticeable peak shifts. This suggests that the functionalisation process has not altered the structure of ceria and that the formation of PDA has occurred primarily on the surface of the nanoparticles. For the PCZ nanoparticles, the XRD peak is shifted towards a higher 2θ value, confirming the formation of PDA over the CZ NPs.

Fig. 3(c) depicts the Raman spectra of CZ NPs. The band at 458 cm^{-1} is due to the pure ceria, which corresponds to the F_{2g} Raman active mode of the fluorite structure [48]. This band arises from the symmetrical stretching vibration of the oxygen atoms around the cerium ions. When Zr^{4+} ions are included in the ceria lattice, the F_{2g} band of pure ceria shifts to approximately 463 cm^{-1} , a higher wavenumber. This shift is accompanied by lattice shrinkage. The observed shift suggests that incorporation of Zr leads to increased structural defects within the mesocrystal units [49]. These structural defects are directly influenced by the positioning of oxygen atoms in the lattice and are sensitive to crystallographic symmetry. In addition, the peak at 608 cm^{-1} is associated with the intrinsic oxygen vacancies generated by Ce^{3+} in the CeO_2 lattices or imperfection resulting from small-size effects [50]. The results indicate that CZ NPs can achieve coherent crystal alignment while retaining a predominant cubic structure even at low Zr^{4+} molar ratios. However, as the Zr^{4+} molar ratio increases, the presence of the tetragonal phase becomes more pronounced alongside the cubic phase. Any physico-chemical contribution with the incorporation of Zr will not result in any discernible shifts to the crystal's shape or structure.

The Raman spectra of P, PC8, and PCZ are presented in Fig. 3(d). A strong band at 460 cm^{-1} and 465 cm^{-1} represents the Ce–O–Ce vibration in PC8 and PCZ NPs, respectively. In Raman spectra, P exhibits two broad characteristic peaks at 1340 cm^{-1} and 1578 cm^{-1} [51]. The synthesised PC8 and PCZ show two relatively weaker bands at 1340 cm^{-1} and 1578 cm^{-1} , which align with the characteristic peaks of P, proving that PDA is successfully synthesised and functionalised on ceria and CZ NPs.

Fig. 3(e) depicts the ATR-FTIR spectra of P, PC8, and PCZ. PC8 and PCZ have a common characteristic peak at approximately 460 cm^{-1} , representing the Ce–O–Ce vibration due to CeO_2 NPs. An intense and broad peak observed around 3240 cm^{-1} is identified corresponding to the hydroxyl (OH) group's stretching vibration [21]. In the spectrum of PDA, absorption bands at around 1596 cm^{-1} and 1506 cm^{-1} correspond to benzene skeleton vibrations [46]. In addition, a peak at 1053 cm^{-1} is also observed because of the vibrational stretching of the C–O bond. A broad peak at $3000\text{--}3600\text{ cm}^{-1}$ is distributed to –NH-based stretching vibrations and catechol –OH stretching vibrations [47]. The FTIR spectra of PC8 and PCZ exhibit distinctive peaks corresponding to CeO_2 and PDA, indicating PDA was successfully functionalised over the surface of ceria and CZ NPs.

The chemical state of the PCZ surface can be interpreted from the detailed analysis of the XPS spectrum shown in Fig. 4. A thorough analysis of the survey spectrum reveals peaks corresponding to Ce 3d, Zr 3d, O 1s, N 1s, and C 1s. However, the peak ascribed to ceria is partially obscured due to a PDA layer overlaying it. The individual peak of every element was deconvoluted using Gaussian approximation.

Multiple oxidation states and the d orbital splitting due to the coupling of spin and angular momentum of 'f' electrons and '3d' core holes make the Ce 3d spectrum complex with six fitted peaks. The peaks named v^1 , v , and v^{11} at lower binding energies correspond to $3d_{5/2}$ splitting of the 3d core orbital of Ce, whereas the peaks at higher energies, namely u^1 , u , and u^{11} , relate to the $3d_{3/2}$ orbit splitting. The final orbital state $3d^{10}4f^1$, because of the Ce^{3+} oxidation state, is responsible for the peaks 'u' and 'v' at 903.2 eV and 884.7 eV , respectively. All other peaks are the indicative of Ce^{4+} oxidation state of ceria [52]. Specifically, two peaks at 182.5 eV and 184.9 eV represent the Zr $3d_{5/2}$ and Zr $3d_{3/2}$ splitting of Zr in a 4+ oxidation state. No peak was observed at 182.9 eV , confirming the absence of ZrO_2 [53,54]. Partial substitution of Ce^{4+} atoms of size $0.97\text{ \AA}$ with much smaller Zr^{4+} ($0.84\text{ \AA}$) creates a lattice contraction in the crystal lattice which is then compensated by the formation of bigger Ce^{3+} ($1.10\text{ \AA}$) atoms leading to a significant presence of peaks corresponding to Ce^{3+} in the XPS spectra of PCZ NPs. This increment in the Ce^{3+} oxidation state is attributed to improved redox activity.

The O 1s peak is deconvoluted to three peaks at 530.6 eV , 532.2 eV , and 534.2 eV , corresponding to lattice oxygen, defective or adsorbed

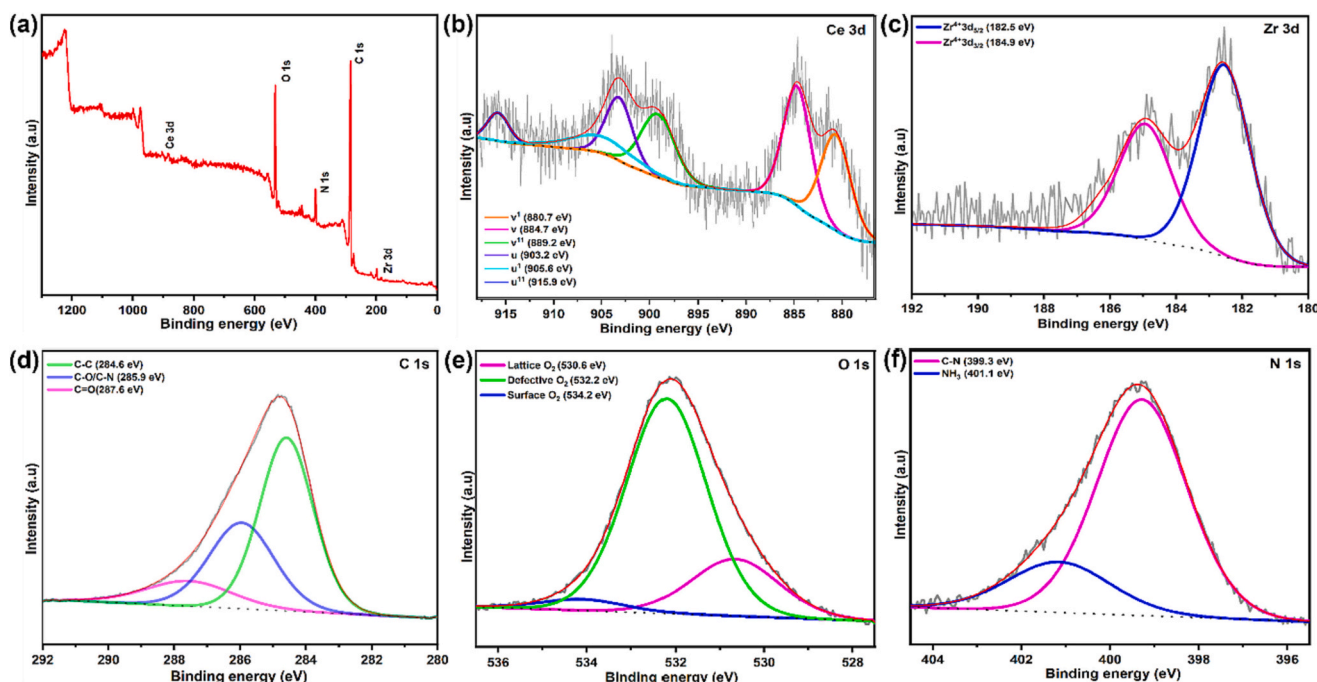


Fig. 4. XPS analysis of PCZ NPs; (a) Survey spectrum. XPS spectra of (b) Ce 3d, (c) Zr 3d, (d) C 1s, (e) O 1s, (f) N 1s.

oxygen (O^{2-}), and surface oxygen species, respectively. In defective oxygen peak and consumption of lattice, oxygen peak indicates more Ce-O-Zr formation, leading to more oxygen vacancies in the material surface. The C—O bond formation and surface oxygen because of PDA functionalisation would also be included in the peaks at 532.2 eV and 534.2 eV [55,56].

The presence of PDA at the surface is prominently indicated by the Carbon (C 1s) and Nitrogen (N 1s) peaks. The C 1s peak at 284.6 eV demonstrates the existence of sp^3 hybridised Carbon in C—C bonds. A peak at 285.9 eV represents Carbon-Nitrogen or Carbon-Oxygen (C—N or C—O) bonds, and the peak substantiates the presence of carbonyl groups (C=O at 287.6 eV). N 1s is fitted with two peaks showing sp^2 hybridised C—N bond because of pyrrolic or indole (R-NH-R) structure in PDA at 399.3 eV and a peak at 401.1 eV is due to the presence of NH_2 [57].

Fig. 5 depicts the variations in OSC values for the concentration of ceria in different CZ NPs as estimated by TGA. As previously stated, including the smaller Zr^{4+} ions in the ceria lattice significantly improves the OSC. This is because of the formation of an imperfect fluorite-structured solid solution, increasing oxygen mobility. Fig. S2 represents the TGA curves of CeO_2 and different CZ nanoparticles. TGA measurements were done in two cycles of heating and cooling, and the weight loss observed during the second heating cycle was considered to evaluate the OSC. The change in oxygen content ($\Delta\delta$) resulting from the change in sample mass during the second heating (Δw) was estimated according to the equation,

$$\Delta\delta = \frac{M_s \cdot \Delta w}{M_o \cdot m_s} \quad (5)$$

where M_s represents the molecular weight of the sample, M_o represents the molecular weight of oxygen, and m_s represents the initial mass of the sample in the air at ambient temperature [58].

It is evident from the graph that all three CZ NPs had higher OSC and improved reducibility compared to pristine ceria. The OSC was highest for the composition CZ4 (0.0392) and lowest for the CZ5 nanoparticles (0.0362). For the CZ2 nanoparticles, the OSC was closer to 0.0381. In addition, XRD analysis of oxygen vacancy concentration also shows a

similar trend for pure ceria and CZ NPs. Due to the structural changes, the CZ5 nanoparticles have lower OSC and oxygen mobility than the CZ4 and CZ2 nanoparticles. The XRD results show that CZ5 has a tetragonal structure, which differs from other CZ NPs (cubic fluorite). The anisotropy of diffusion in a tetragonal structure hinders the movement of oxygen, as the varied orientation of grain boundaries limits it. This leads to a lower overall diffusion rate in polycrystalline materials. However, in the cubic fluorite structure, the introduction of smaller Zr^{4+} ions results in a decrease in unit cell volume and induces stress. This stress promotes the formation of crystal imperfections, which facilitates high oxygen diffusion [59].

The nanocomposite coatings were prepared using the spin-coating method and the coating thickness was measured to be $10 \pm 5 \mu m$. Reducing the thickness of the coating mitigates the peeling behaviour of the coating to an extent. The roughness of the coatings for E and EPCZ coating was determined to be $22 \pm 4 nm$ and $14 \pm 3 nm$, respectively. A slight decrease in the average roughness was observed with the addition of PCZ NPs. Thus, EPCZ coatings create a smooth surface that can inhibit microbial adhesion. The surface morphology of the pure epoxy and EPCZ coatings was examined from FESEM images, which are given in Fig. S3 (see supplementary information). The presence of nanosized pores in the pure epoxy coating allows for the easy passage of water and other electrolyte media, leading to corrosion of the metal over time. However, the addition of PCZ nanoparticles effectively fills these pores, thereby preventing the diffusion of the corrosive media. The SEM images of the epoxy coating reveal the presence of pores (Fig. S3a,b), while that of the EPCZ coating exhibits a smoother surface with a significantly reduced number of pore density (Fig. S3c,d). This shows that PCZ nanoparticles are present in the epoxy matrix. The EDAX spectra of pure epoxy and EPCZ coatings are depicted in Figs. S4 and S5, respectively. The C, O and N peaks are present in epoxy coatings, whereas additional peaks of Ce and Zr were observed in EPCZ coatings. This confirms the incorporation of nanoparticles in epoxy coatings. The elemental composition of the coatings is shown in Tables S1 and S2. The FESEM cross sectional and top view images of EPCZ coatings are shown in Fig. S6(a) and (b) respectively. It was observed that the obtained thickness of the EPCZ coatings from the cross-sectional image is

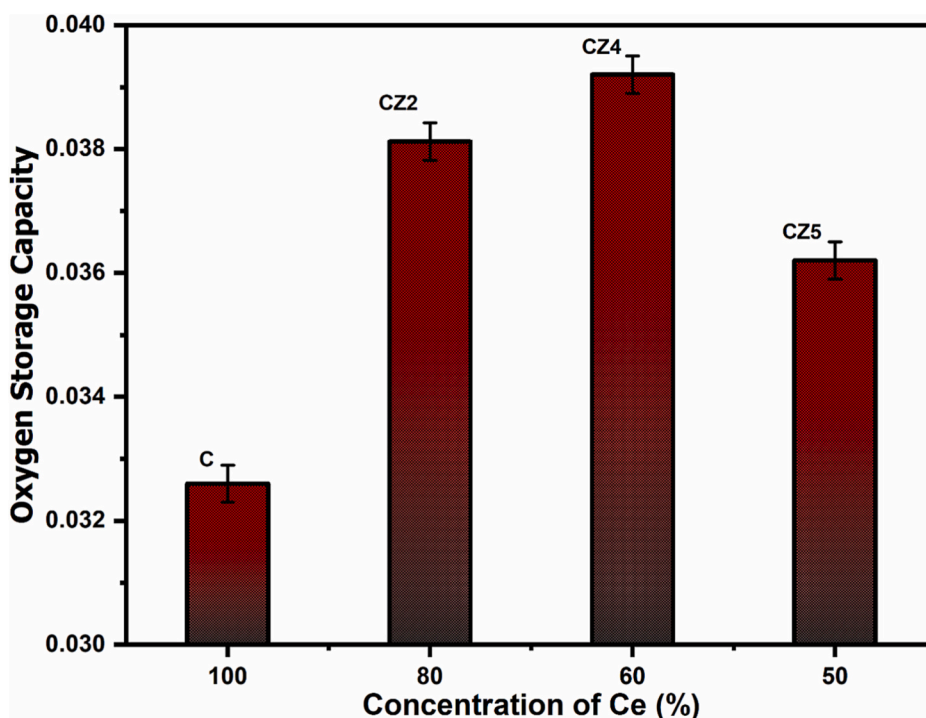


Fig. 5. Oxygen storage capacity of ceria, CZ2, CZ4 and CZ5 nanoparticles.

consistent with the thickness value obtained using Surfex pro gauge. The uniform distribution of PCZ nanoparticles (Ce, Zr, C, N, and O) in the epoxy matrix are also shown in the Fig. S6(c–g).

3.2. Antibacterial studies

The antibacterial activity of synthesised CeO₂, PDA functionalised CeO₂, CZ and PCZ NPs towards gram +ve and gram –ve bacteria were examined, and the number of CFUs was obtained for the concentration of 1 mg/mL. It was evident from Fig. 6 that the number of CFUs produced by each bacterial strain decreases significantly compared to the positive control, indicating that these nanoparticles inhibit the growth of bacteria effectively. The direct and indirect interactions between CeO₂ NPs and the bacterial cell membrane surface cause cell membrane destruction. The variation in CFUs is influenced by factors such as the kind of bacteria, nanoparticle concentration, shape, size and surface area. The changes in the structure and configuration of cell membrane might also account for the differences in action against the variety of bacteria tested here. The reduced number of CFUs, indicates that gram –ve bacteria (Fig. 6a,c) are more susceptible to these nanoparticles than gram +ve bacteria (Fig. 6b).

The gram +ve bacteria have a thick peptidoglycan-coated cell wall. In contrast, gram –ve bacteria possess an outer lipopolysaccharide membrane that can trigger an immune response and enhance nanoparticle susceptibility. The increased permeability of the external membrane of gram –ve bacteria allows the nanoparticles to penetrate more easily and disrupt essential cellular processes. The ceria nanoparticles get electrostatically attracted to the external membrane of the gram –ve organisms [60]. The adsorption to the membrane may cause

membrane disruption, exposing the inner membrane and the periplasmic space, making these organisms more susceptible to the nanoparticle. The presence of ceria nanoparticle on gram-negative organisms' surfaces has proven to generate ROS and intrusion of the nutrient transport chain [61,62]. Figs. S7, S8 and S9 represent the number of CFUs of *E. coli*, *S. aureus* and *P. aeruginosa* for different nanoparticles, respectively. It is evident from the figures that CFUs for PC4 treated samples are higher than PC8 treated samples. PC8 has more ceria concentration than PC4; thus, better antibacterial activity is observed. Addition of Zr improves the OSC of CZ NPs, out of which CZ4 has the highest value. The observation of lower CFUs for the CZ4 treated sample in the figures validates the previous observation (Figs. S7, S8 and S9). In general, PCZ NPs exhibited a greater antibacterial effect than control, as evidenced by the reduction of CFUs by 82 % in the case of *P. aeruginosa* and 83 % for *S. aureus*.

The antibacterial activity of EC, EPC8, ECZ4 and EPCZ nanocomposites was also studied, as shown in Fig. 7. All the experiments were conducted in triplicates. In comparison to the positive control, EPCZ composites reduced CFUs by 93 % for *S. aureus* and 87 % for *P. aeruginosa*. This shows a remarkable antibacterial effect for all bacterial strains. The combined effect of water-borne epoxy composites and PDA allowed greater dispersion of ceria and CZ NPs to enhance nanocomposites' antibacterial activity.

3.3. Antibiofilm efficiency of polydopamine functionalised ceria-zirconia nanoparticles

A semi-quantitative crystal violet staining analysis was adopted to analyse the effect of nanoparticles on biofilm formation. The antibiofilm

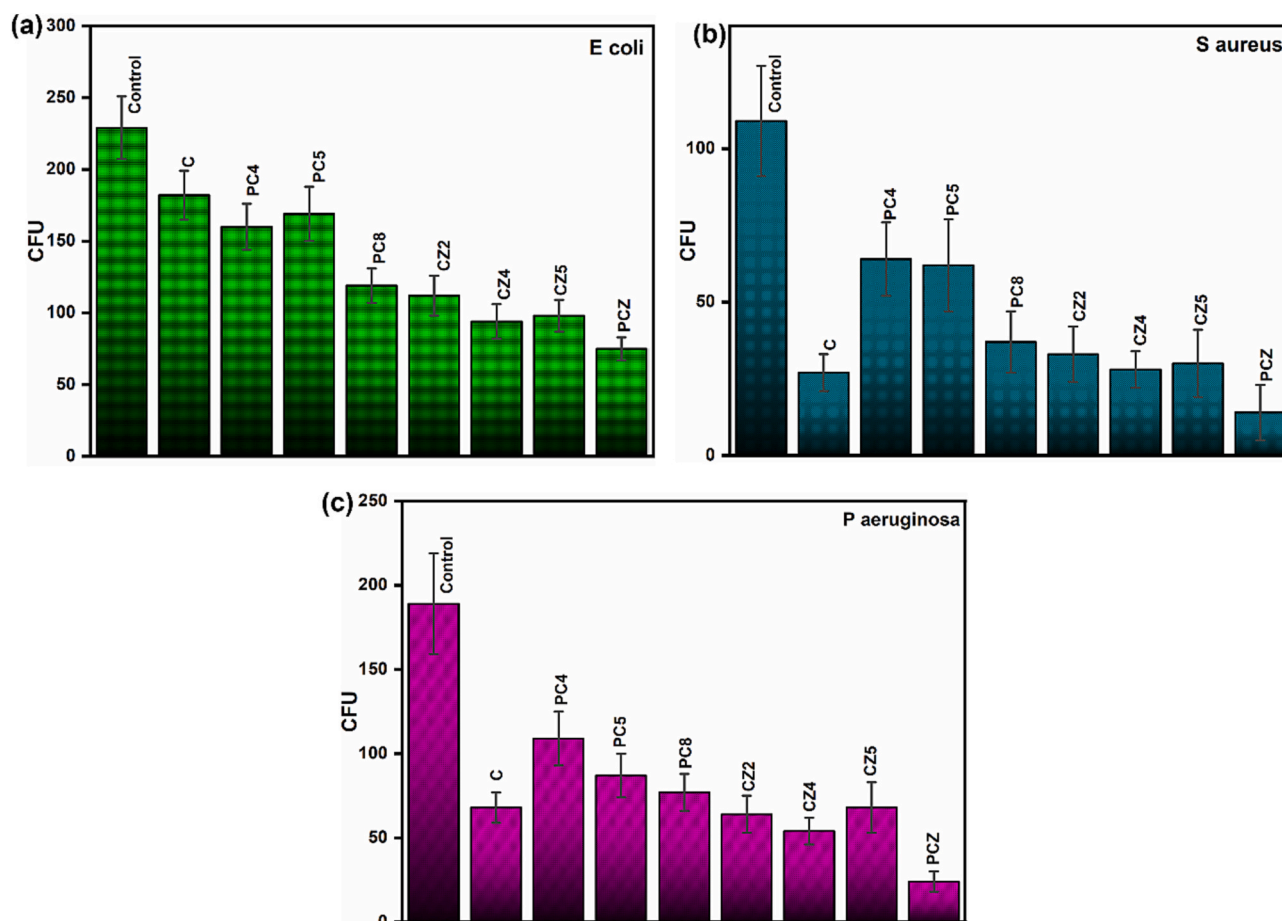


Fig. 6. Antibacterial activity of nanoparticles against (a) *E. coli*, (b) *S. aureus* and (c) *P. aeruginosa*.

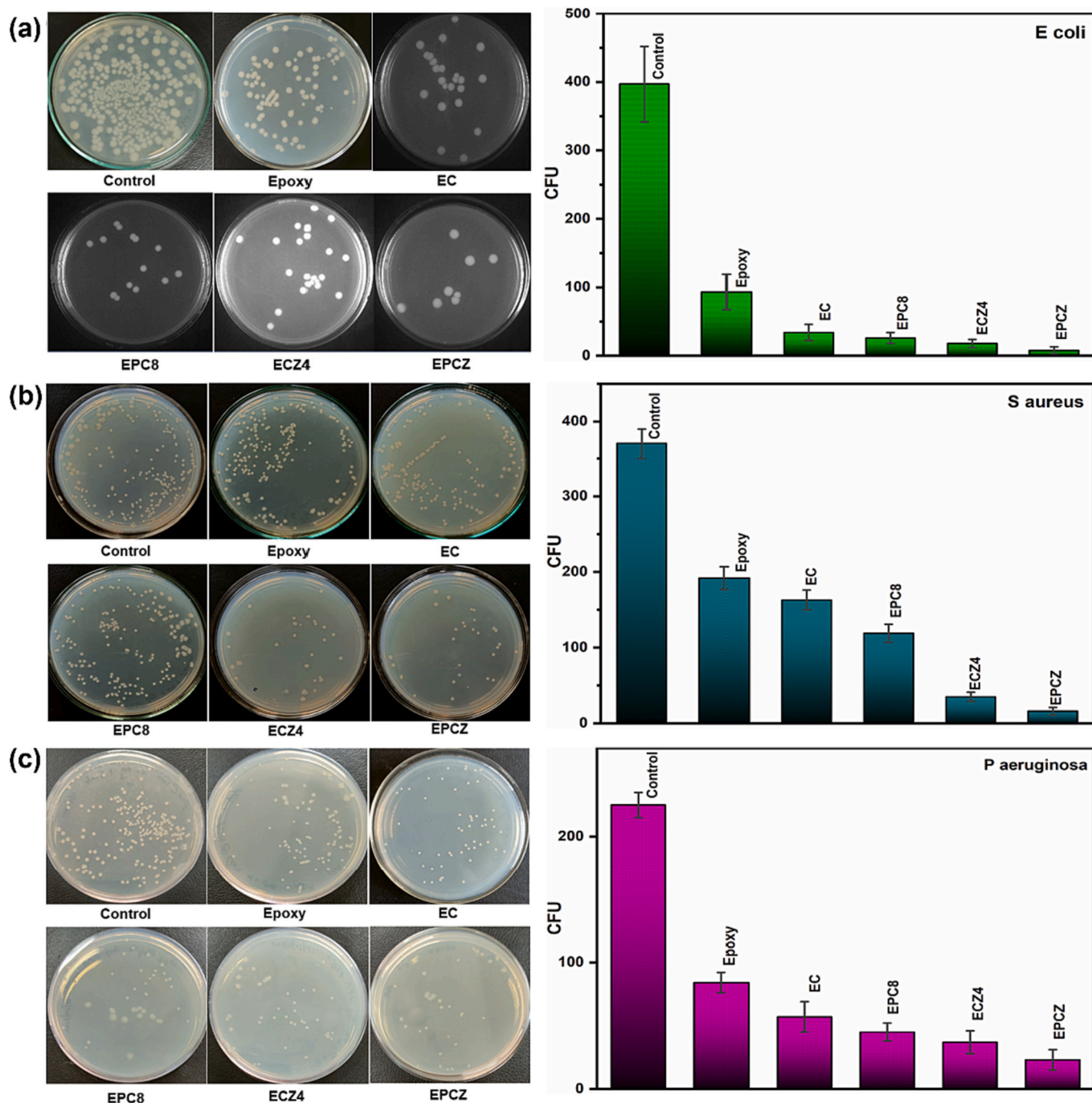


Fig. 7. Antibacterial activity of epoxy and different nanocomposite coatings against (a) *E. coli*, (b) *S. aureus* and (c) *P. aeruginosa*.

investigations on C, PC8, CZ4, and PCZ obtained from the biofilm assays revealed that each NP exhibited distinct inhibitory action against *P. aeruginosa* and *S. aureus*. Fig. 8(a) shows that all coatings with nanoparticles exhibit a concentration-dependent biofilm inhibition towards *P. aeruginosa*. Interestingly, it was found that for a concentration of 400 $\mu\text{g/mL}$, PCZ nanoparticles showed superior antibiofilm activity than C, PC8 and CZ4 mixed oxide nanoparticles against gram-negative bacteria. PCZ nanoparticles exhibited a significant reduction in biofilm destruction, with suppression levels reaching up to 95 % compared to control at this particular concentration. The biofilm inhibition efficiency against *P. aeruginosa* was estimated and plotted in Fig. 8(d). The results clearly show that the highest efficiency in inhibiting biofilm formation was achieved for a concentration of 400 $\mu\text{g/mL}$ for every sample and was

found to be 35.91 %, 57.09 %, 71.10 % and 96.37 % for C, PC8, CZ4 and PCZ nanoparticles, respectively. This is attributed to the synergistic antibacterial effect of ceria nanoparticles, functionalisation with PDA and inclusion of Zr in ceria, which significantly improved antibiofilm activity, resulting in biofilm destruction.

Fig. 8(b) shows biofilm inhibition by C, PC8, CZ4 and PCZ nanoparticles against *S. aureus* for different concentrations, up to 400 $\mu\text{g/mL}$. The biofilm inhibition efficiency was also estimated (Fig. 8(c)) and maximum efficiency was observed for a concentration of 400 $\mu\text{g/mL}$ for all the samples. At a 400 $\mu\text{g/mL}$ concentration, C, PC8, CZ4 and PCZ nanoparticles inhibited biofilm formation in *S. aureus* by 15.99 %, 44.23 %, 51.95 %, and 62.67 %, respectively. This indicates that even at low nanoparticle concentrations (400 $\mu\text{g/mL}$), PCZ nanoparticles showed

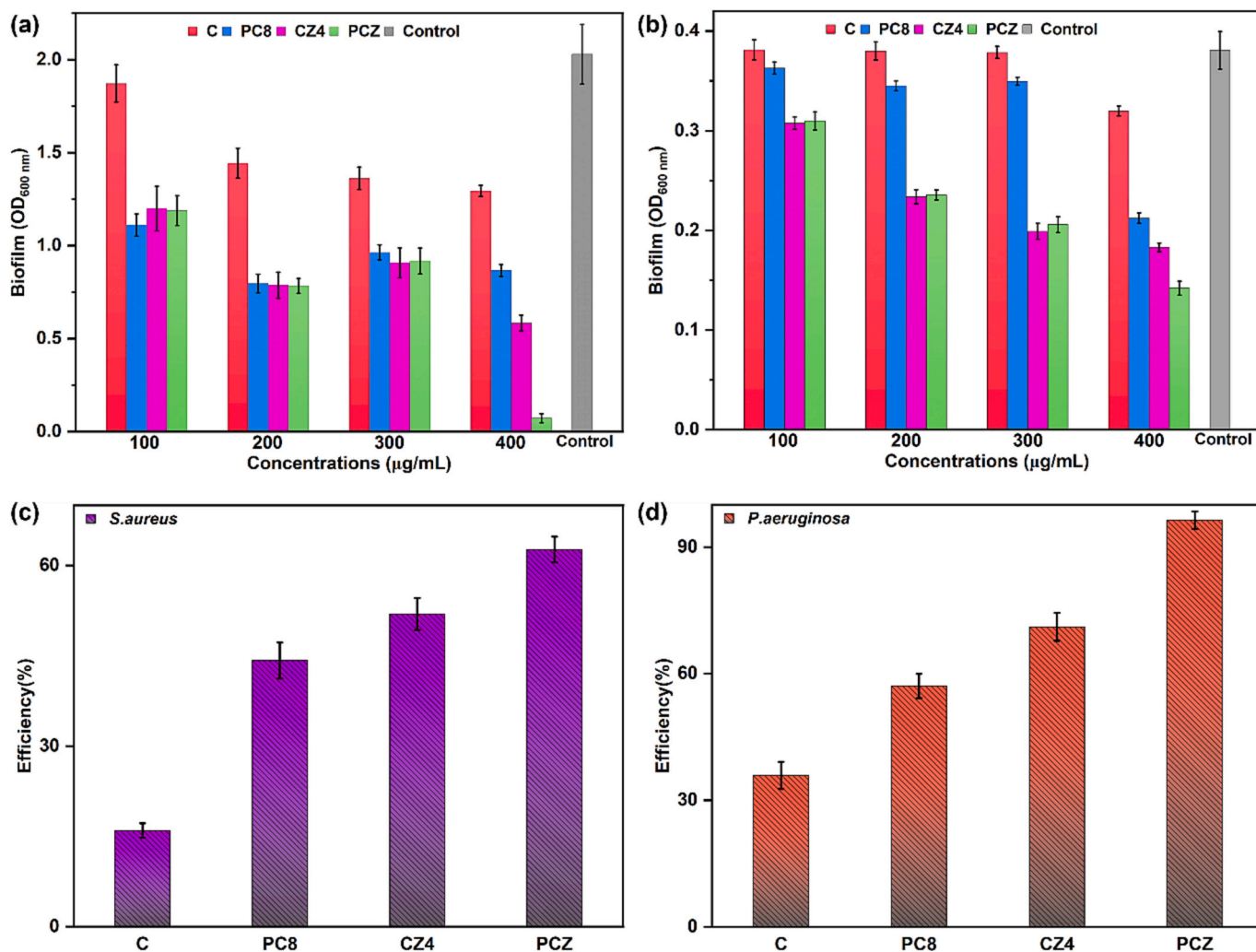


Fig. 8. Antibiofilm activity of different nanoparticles against (a) *P. aeruginosa* (b) *S. aureus* and the biofilm inhibition efficiency of different nanoparticles against (c) *S. aureus* (d) *P. aeruginosa*.

high biofilm inhibition. However, the disparity in biofilm inhibition between PCZ and C nanoparticles was more significant in *P. aeruginosa* (96.37 %) than in *S. aureus* (62.67 %).

The nanoparticles had a stronger antibiofilm impact on *P. aeruginosa* compared to *S. aureus*, which is attributed to the variation in bacterial structure. Gram -ve bacteria have a thin peptidoglycan cell wall surrounded by a lipopolysaccharide-containing outer membrane. In contrast, gram +ve bacteria is attributed with multilayered peptidoglycan, making it difficult for PCZ nanoparticles to damage their cell wall for subsequent biofilm inhibition [62].

3.4. Biofilm inhibition studies

SEM analysis of bacterial biofilms was carried out to examine the effect of E, EC, EPC8, ECZ4, and EPCZ nanocomposites on the biofilm architecture of the test bacterium. Fig. 9 depicts SEM images of the coating surfaces immersed in the inoculated medium for 5 days containing approximately 10^8 CFUs of *P. aeruginosa*. In the absence of antibacterial agents (control), a significant number of *P. aeruginosa* get adhered to the surface and forms dense biofilms surrounded by the extracellular matrix (Fig. 9(a)). After treatment with nanocomposites, the biofilms of *P. aeruginosa* were not observed, and only small bacterial clusters were adhered to the coating's surface (EC, EPC8, ECZ4), as shown in Fig. 9(c-e). Intriguingly, EPCZ coatings (Fig. 9(f)) exhibit the greatest biofilm inhibition compared to pure epoxy coatings (Fig. 9(b)),

validating the result obtained by the crystal violet staining technique. Enhancement of biofilm inhibition by EPCZ coatings could potentially be achieved through the promotion of cellular toxicity and oxidative stress, facilitated by the release of Ce^{3+} ions and significant numbers of oxygen vacancies. These factors contributed to the generation of ROS and resulted in cell death [47].

Similarly, the formation of biofilm by *S. aureus* was analysed using SEM on E and EPCZ coating surfaces (Fig. S10). The results indicated that EPCZ coatings exhibited the strongest antibiofilm effect compared to pure epoxy coating and positive control. The plausible reasons for the increased antibiofilm activity of these nanocomposite coatings could be (i) the intrinsic antibacterial activity of ceria nanoparticles due to its reversible oxidation states, (ii) Zr doping increases the OSC in the mixed oxide nanoparticles, and (iii) increased dispersibility of CZ4 nanoparticles in the epoxy matrix due to the surface functionalisation with PDA. Thus, SEM analysis revealed that the nanocomposite coatings successfully inhibit biofilm formation, protecting coatings and mild steel from microbe-induced failure.

In order to comprehend the efficiency of nanocomposite coatings in real-world scenarios, seawater consortiums was utilised in the present work. The bacterial adhesion and anti-biofouling activity on control, pristine epoxy, and EPCZ coatings were assessed via SEM imaging, subsequent to a 15-day immersion in a nutrient broth medium inoculated with bacteria. The thick bacterial cells and extracellular polymeric substances covering the coating surface are visible in the SEM image for

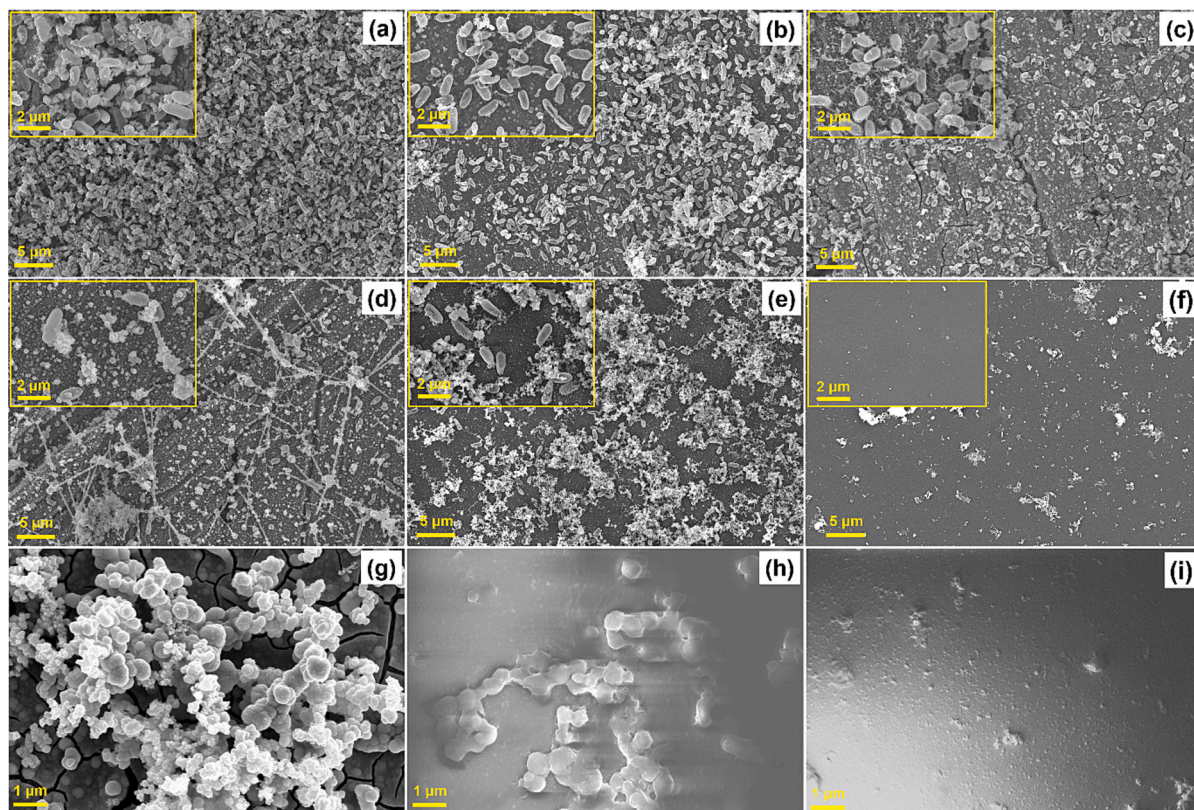


Fig. 9. (a–f) SEM images of uncoated control, epoxy coated and composites coated mild steel surface after treating with bacteria ((a) Control, (b) Epoxy (c) EC (d) EPC8 (e) ECZ4 and (f) EPCZ). (g–i) SEM images of uncoated control, epoxy coated and EPCZ coated mild steel surface after treating with sea water consortium ((a) Control (b) Pure epoxy (c) EPCZ).

the control (Fig. 9(g)). For pure epoxy coatings, tiny clusters and microcolonies of bacterial organisms can be seen (Fig. 9(h)), whereas EPCZ coatings exhibit strong antibiofilm capability (Fig. 9(i)). Thus, SEM images indicate that the EPCZ coating surface has significant anti-biofouling activity.

A comparison between the present work and recent studies on anti-biofouling nanocomposite polymer coatings is shown in Table 3. The results indicate that EPCZ coatings have superior effectiveness in inhibiting biofilm formation, despite having lower nanofiller content and thinner coated film.

3.5. Anti-biocorrosion studies

EIS is an effective technique for studying the anti-biocorrosion protection of coatings, and it is a non-destructive testing method. The Nyquist and Bode plots were used to assess the anti-biocorrosion properties of different coating systems (E, EC, EPC8, ECZ4 and EPCZ) after

immersing in NB containing *P. aeruginosa* for varying immersion times (12, 48, and 84 h). The results are shown in Fig. 10. The biocorrosion resistance of protective coating films can be assessed based on the semicircle shape and its radius observed in the Nyquist plots [15]. A larger radius of the semicircle signifies higher level of resistance to biocorrosion, signifying enhanced efficacy of the coating film in preventing corrosion induced by biological factors. The Nyquist plot (Fig. 10(a–c)) reveals that EPCZ coatings have the largest capacitive impedance arc diameters, followed by E, EC, EPC8, and ECZ4 coatings. Fig. 10(d–f) depicts the Bode plots providing valuable information on the behaviour of the coating in a corrosion-prone atmosphere. These plots are divided into two sections: the higher frequency region reflects the impedance at the coating surface, while the lower frequency region represents the charge transfer process at the interface between the coating and the metal substrate [17]. Generally, the impedance value observed at the lowest applied frequency (0.1 Hz) could be described as a measure of the protective property of the respective coating. In Fig. 10

Table 3

Recent studies on antibiofouling coatings over metal substrates.

Polymer matrix	Nanofiller	Nanofiller concentration (wt%)	Film thickness (μm)	Observations	Ref:
PDMS	Fluorinated MWCNTs	0.05, 0.1 and 0.2	175 ± 25	Improved fouling release properties by 67 %	[63]
Epoxy	APTES- ZnO	2.5	100	Excellent anti-biofouling and anticorrosion activity	[64]
PDMS	SiO ₂ -doped ZnO NPs	0.5	150	Minimum fouling adhesion	[65]
Silicon Rubber	GO-Al ₂ O ₃	1	150	Maximum superhydrophobicity and minimum microbial adhesion	[66]
Water-borne Epoxy	PANI-GO	12	200 ± 15	High anti-biofouling and corrosion performance	[67]
Water-borne Epoxy	PCZ	0.5	10 ± 5	Superior biofilm inhibition by 96 % against <i>P. aeruginosa</i>	Present work*

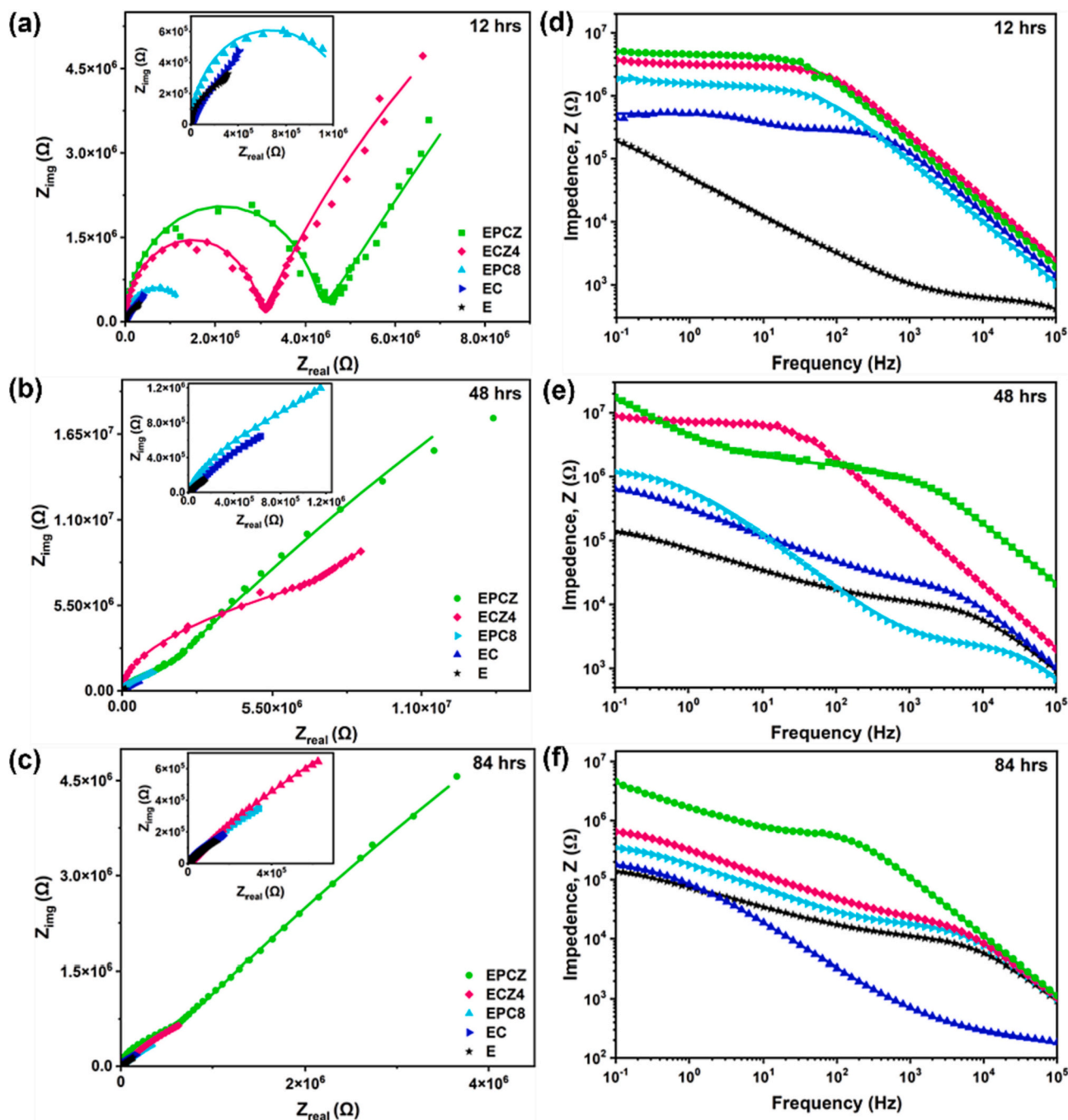


Fig. 10. Anti-biocorrosion studies using EIS: (a–c) Nyquist plot obtained after prolonged immersion of EPCZ coated mild steel substrate ((a) 12, (b) 48, and (c) 84 h). The inset depicts the enlarged view at lower impedance. (d–f) The corresponding Bode plots displaying the impedance modulus at different immersion times ((d) 12, (e) 48, and (f) 84 h).

(d–f), shows an increased immersion time from 12 to 84 h, and the impedance value of the epoxy coating reduced from approximately $2.0 \times 10^5 \Omega$ to $1.37 \times 10^5 \Omega$, indicating that the epoxy coating rapidly loses its anti-biocorrosion resistance. For the EC coating, the impedance value increases to $5.32 \times 10^5 \Omega$ after 12 h of immersion. The impedance value was found to increase to $6.4 \times 10^5 \Omega$ after 48 h, followed by a decrease to $1.8 \times 10^5 \Omega$ at the end of 84 h. This observation indicates the possibility that addition of ceria nanoparticles to the epoxy resin coating will increase its resistance to bio-corrosion. However, the abrupt decrease in the impedance of EC coating may result from the inadequate dispersion of ceria nanoparticles within the coating matrix.

Furthermore, severe aggregations result in numerous through-pore defects in the epoxy coating, which facilitates the ingress of aggressive ions. Adding PDA as a functionalising agent to ceria nanoparticles enhances their dispersibility in the epoxy coating, resulting in improved biocorrosion resistance. It was shown that even after 84 h of immersion, the impedance value of EPC8 coatings is $3.4 \times 10^5 \Omega$, indicating that EPC8 coatings have superior anti-biocorrosion performance compared to EC coating. This improved performance could be due to the presence of nano-fillers within the EPC8 coatings, which effectively restrict the ingress of water and aggressive ions, thereby inhibiting bacterial growth. ECZ4 coatings exhibited greater biocorrosion resistance than EC

coatings and EPC8 coatings. Even after 84 h, the impedance value of ECZ4 coating is approximately $6.5 \times 10^5 \Omega$. CZ NPs have a higher impedance because nanoparticles with smaller crystallite size occupy more pore areas, thereby increasing the impedance of electrolyte diffusion. The EPCZ coatings' impedance values remain consistently higher ($4.7 \times 10^6 \Omega$) throughout the immersion period than all other coatings in the low-frequency region.

In contrast, as the immersion time increases, the total impedance decreases exponentially for all coatings except the EPCZ coating. In addition, the rate of impedance reduction is minimal for the EPCZ coating. This is due to the fact that PDA functionalization improves CZ4 NP dispersion in epoxy resin, thus preventing the aggregation of nanoparticles. In addition, the presence of zirconia in CZ NPs increases the OSC, leading to reduced bacterial adhesion and biofilm deposition on the underlying mild steel substrate.

The semi-circular arc in the Nyquist plot represents the capacitive behaviour of the coatings. Multiple semicircles in both high frequency and low-frequency regions suggest that the corrosion process involves multiple time constants. The diameter of impedance curves in the Nyquist plot increases in the presence of coatings. Among the coatings studied, the EPCZ nanocomposite exhibits the largest diameter, followed by ECZ4, EPC8, EC, and pure epoxy. This is primarily due to the combined effect of PDA and CZ nanoparticles, which improve the barrier properties of the EPCZ nanocomposite coating. Moreover, the impedance value of the EPCZ coating does not change substantially throughout the immersion process, which further demonstrates that the EPCZ coating adheres well to the metal substrate.

The impedance values obtained from the Nyquist plot were analysed using various equivalent circuits, and the circuits that best fit the data are depicted in Fig. 11. The corrosion deterioration process of the coatings can be conceptualised as multiple electrical equivalent circuits with distinct time constants. Different circuits were employed to fit the EIS data since the biocorrosion protection mechanism will undergo significant changes during the immersion period. The derived data was analysed to determine the effect of microbes on the electrode-electrolyte interface so that biofilm formation can be understood. The electrical analogue circuits consist of solution resistance (R_s), coating pore resistance (R_c), coating capacitance (C_c), double layer capacitance (C_{dl}), and charge transfer resistance (R_{ct}) and Warburg diffusion (Z_w) [21,68]. In all the fittings, constant phase elements Q_c and Q_{dl} were introduced to replace the capacitances C_c and C_{dl} because of the deviations from ideal capacitive behaviour [35]. At different immersion times, the EIS data for all coatings were fitted using the equivalent circuit of R_s , Q_c , R_c , Q_{dl} , and R_{ct} . The fitted data and circuit parameters obtained from analysing the Nyquist plot of different coatings can be found in Table S3. The circuit consists of two distinct time constants (Q_c R_c and Q_{dl} R_{ct}) representing two electrode processes occurring at the coating and the metal-coating

interface. Prior to diffusing through the coating, the electrolyte encounters a small resistance referred to as the solution resistance (R_s). The R_c and R_{ct} are crucial parameters for assessing the protective capability of the metal, where R_c indicates the degree of defects and deterioration in epoxy coatings. As the coating deteriorates, more defects will appear, resulting in a reduction in coating resistance.

Additionally, R_{ct} is proportional to the corrosion rate at the interface between the electrolyte and the metal. A higher R_{ct} indicates a lower corrosion rate. The constant phase element Q_c corresponds to the capacitance of the coating, indicating its ability to absorb water. On the other hand, Q_{dl} represents the double-layer capacitance, signifying the metal's active surface area that is susceptible to corrosion. In the Nyquist plot, the nanoparticles incorporated epoxy coatings show a linear region instead of a capacitive loop. This linear behaviour corresponds to the Warburg (Z_w) impedance with a constant phase angle of 45° , indicating the control of the diffusion of corrosive species at the electrode/electrolyte interface.

Electrochemical reactions occur at the electrode surface after a prolonged immersion in the electrolyte, forming corrosion products at the metal/coating interface. Since pure epoxy coatings are porous, they can form an intermediate oxide layer, promoting corrosion. On the other hand, the corrosion product layer formed in the EPCZ coating consists of hydroxides of the Ce, which precipitate onto the metal surface. These hydroxides have low solubility and serve as a protective barrier to prevent the penetration of corrosive ions. As a result, the circuit needs an additional time constant to fit the EIS data [69]. The new time constant (Q_{ox} & R_{ox}) is shown in Fig. 11 where Q_{ox} denotes the oxide layer capacitance, and R_{ox} denotes oxide layer resistance, respectively. The evolution of oxide layer resistance (R_{ox}) and capacitance (Q_{ox}) is observed for the EPCZ coatings after prolonged immersion times. The EPCZ coatings are found to have a greater R_{ox} value ($89.69 \times 10^6 \Omega$) at 48 h immersion times. The capacitance values of the oxide layer provided additional evidence for formation of a passive film. In addition, the EPCZ coatings exhibit a decrease in Q_{ox} after 48 h of immersion, implying the formation of a stable oxide/hydroxide film. This film restricts the contact between the electrolyte and the coating, diminishing the active contact area and decreasing Q_{ox} .

It is evident from Table S3 that the R_{ct} value of the EPCZ coatings is greater after 12 h of immersion. After that, R_{ct} demonstrated a significant decline for the same coatings. The EPCZ coating maintained a constant and high R_{ct} value with a slower reduction rate, whereas the R_{ct} values of all other coatings decreased sharply as immersion time increased. This could be due to the improved barrier properties of the epoxy coating incorporated with nanoparticles, which facilitates the formation of a stable passive layer that inhibits the transfer of charges across the interface. Thus, by inhibiting bacterial adhesion and biofilm deposition, EPCZ coatings provide a higher biocorrosion resistance in the underlying steel.

Thus, it is concluded that EPCZ coatings have superior corrosion resistance than pure epoxy coatings, even after prolonged immersion. The loose layer in epoxy coatings can accelerate corrosion, creating a synergistic effect between bacteria and aggressive corrosive anions, leading to a deterioration in coating performance. However, PCZ nanoparticle-filled epoxy coatings showed high resistance to degradation, ascribed to the formation of passive film composed of stable oxide/hydroxide at the metal-coating interface. This passive film serves as a shield against the penetration of corrosive ions and also inhibits microbial growth due to the antibacterial properties of PCZ nanoparticles. These findings align with the antibiofilm activity trend observed in the SEM studies.

3.6. Mechanism for anti-biofouling activity

A possible anti-biofouling mechanism of water-borne epoxy coatings incorporating PCZ NPs likely involves direct and indirect contact processes (Fig. 12). In direct contact, positively charged ceria nanoparticles

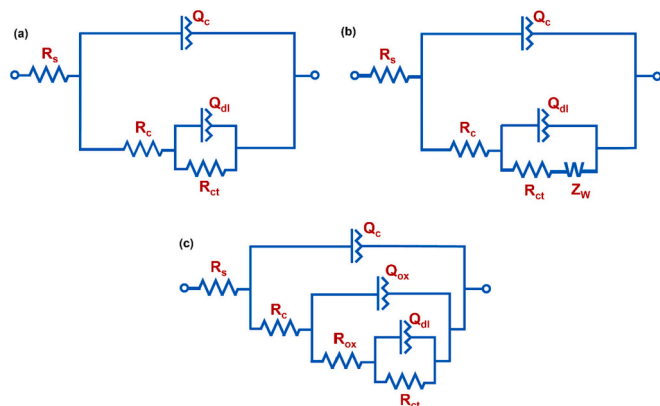


Fig. 11. Electrical equivalent circuit with (a, b) two-time constants and (c) three-time constants.

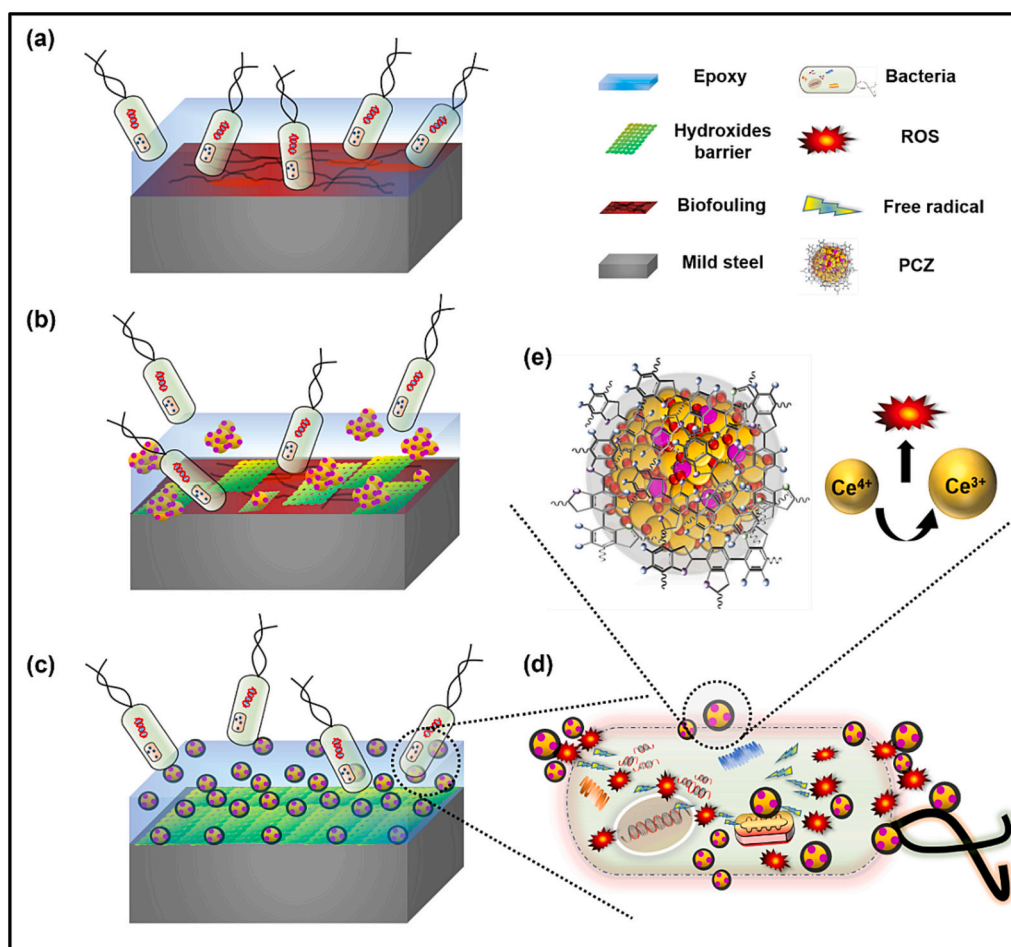


Fig. 12. A scheme showing the plausible ways in which bacteria effects the mild steel coated with (a) pure epoxy, (b) ECZ and (c) EPCZ coatings. 12 (d) an enlarged view for the diffusion of nanoparticles through the bacteria cell wall and its destruction. 12 (e) Scheme for the structure of PCZ and mechanism of ROS generation.

adsorb efficiently on bacterial cells due to the electrostatic interaction. As a result of this interaction, the cellular proteins become non-functional, allowing the nanoparticles to enter the bacterial cell. Once nanoparticles enter the cells, enzymes are disabled, leading to the generation of hydrogen peroxide (ROS) and other free radicals, which further harm the bacterial DNA, RNA, and protein, resulting in microbial cell death. While in indirect contact, CeO₂ nanoparticles undergo a reduction process where Ce⁴⁺ ions are converted to Ce³⁺ ions by releasing superoxide ions (ROS). This reduction process creates oxygen vacancies within the lattice structure as evident from the XRD. The superoxide ions can permeate the bacterial cell membrane, causing damage to nucleic acids and proteins. Eventually, this damage leads to the death of the bacterial cells [14,15]. It is important to note that the existence of oxygen vacancies and active sites can modify the biological and optical characteristics of CeO₂ nanoparticles.

The addition of zirconia with ceria nanoparticles to obtain mixed oxide nanoparticles improves the redox activity, generating more ROS. The TGA results indicate an improvement in OSC for CZ NPs, compared to the pristine ceria nanoparticles, thereby increasing the formation of cations. Some Ce⁴⁺ and Ce³⁺ cations escape from the surface and react with OH groups in water, forming a hydroxide barrier over the metal surface as per the equations below. The hydroxide barrier containing insoluble precipitates of Ce(OH)₄ and Ce(OH)₃ restricts the bacteria from cruising through the coating [17].



Polydopamine is a common biopolymer used for effective surface functionalisation of nanoparticles with antibacterial properties. The nitrogen and oxygen-based surface groups on the PDA chain creates a strong bond with CZ NPs and restricts their agglomeration. The quinone and hydroxyl quinone side chains promote the formation of a framework inside the epoxy resin to improve the dispersibility of nanoparticles. PDA also aids ROS generation due to catechol moieties in its polymer network [35,47]. Apart from providing a coating platform, water-borne epoxy resin creates a thin hydration layer over the coating due to the hydroxyl groups on its surface, which acts as a physical barrier for bacterial adhesion [70,71]. The combined effects of CZ NPs, PDA functionalisation and epoxy matrix make the as-prepared nanocomposite coating suitable for anti-biofouling applications.

4. Conclusions

The present work investigated the enhanced anti-biofouling property of water-borne epoxy coating by the incorporation of PDA-functionalized ceria-zirconia nanoparticles. The PCZ NPs and their control samples were prepared by co-precipitation and chemical functionalization. XRD, Raman spectra, FTIR and XPS techniques confirmed the Zr incorporation in ceria as well as PDA functionalization. The incorporation of Zr into CZ NPs increased their oxygen storage capacity, as determined by TGA analyses. This improvement in oxygen storage

capacity contributes to the enhanced anti-biofouling activity of the NPs. The PDA functionalization enhanced the crosslinking and dispersibility of NPs within the epoxy matrix. The antibacterial property of PCZ NPs was found to be superior compared to other control samples. Antibiofilm studies by crystal violet staining method show that PCZ nanoparticles inhibit biofilm formation by 96 % against *P. aeruginosa* for a concentration of 400 µg/mL, and by 63 % against *S. aureus*. SEM analysis of EPCZ coatings show that addition of PCZ NPs into the epoxy resin coating significantly improves antibiofilm activity, as compared to the control coatings. Furthermore, the prolonged EIS analysis demonstrates a significant enhancement in charge transfer resistance and coating resistance for the EPCZ coating. The EPCZ coatings function as a protective barrier, preventing the penetration of corrosive ions and inhibiting bacterial growth. This, in turn, prevents the degradation of the coated surface and enhances its corrosion and biofouling protection performance. The addition of Zr in ceria enhances the ROS species generation and oxygen storage capacity of the NPs that led to bacterial cell damage. Moreover, the presence of PDA also contributed to antibiofilm activity due to the ROS generation from its catechol moieties. The enhanced antibiofouling properties and stability of EPCZ coatings make them highly suitable for widespread applications in the marine industry and other related disciplines.

CRedit authorship contribution statement

Athira S. Vijayan: Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Visualization. **Antony Joseph:** Investigation, Resources. **Abey Joseph:** Investigation, Resources. **T.S. Abhijith:** Investigation, Resources. **Baiju G. Nair:** Writing – review & editing, Supervision. **V. Sajith:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that there is no conflict of interest.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.108094>.

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